

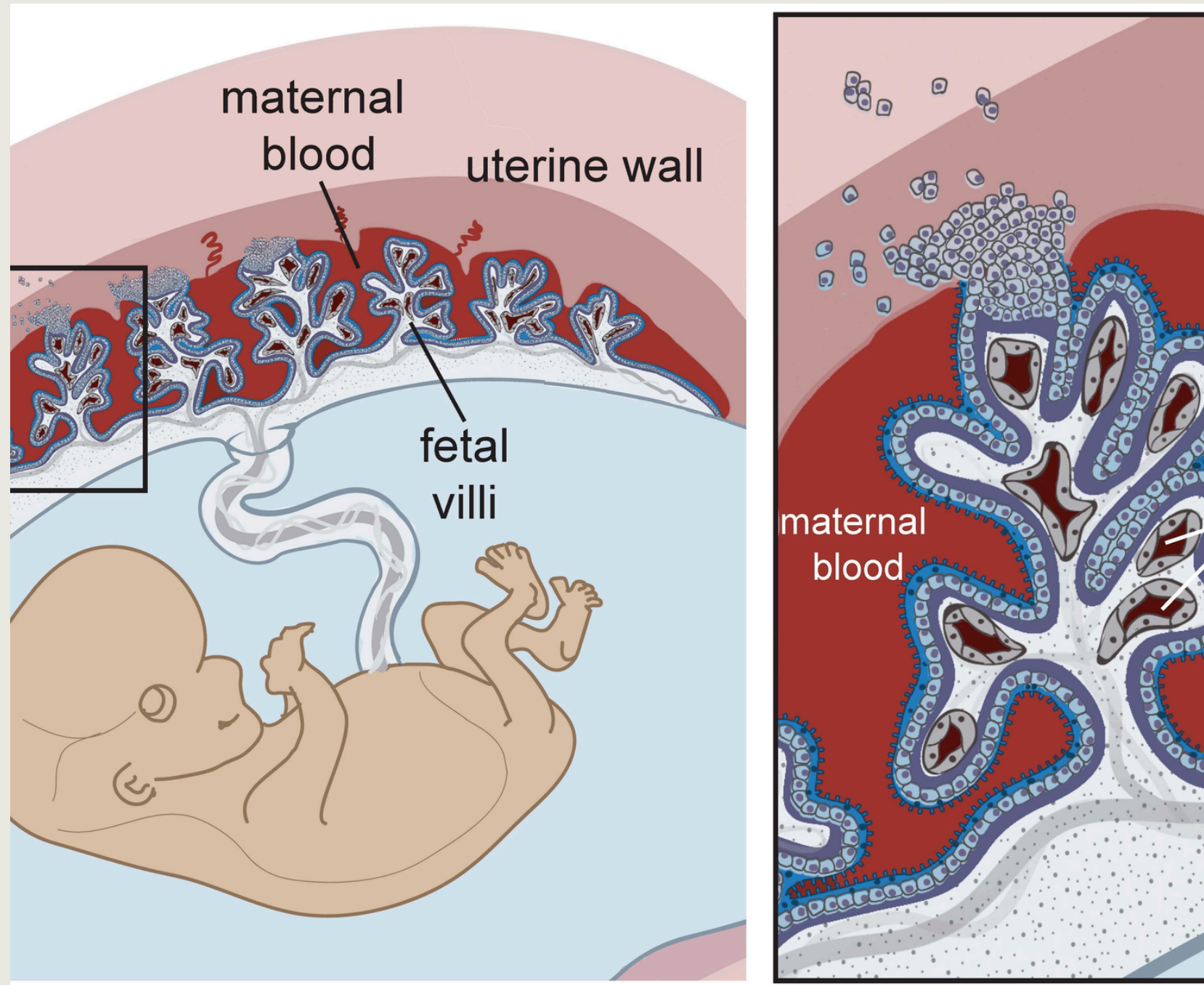
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Maternal Smoking and Preterm Delivery

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Introduction



Numerous studies have shown an increased risk of preterm delivery among women who smoke during pregnancy.

The biological mechanisms underlying the association between smoking and preterm delivery are complex.

- Smoking is known to affect placental function, leading to impaired oxygen and nutrient supply to developing fetus.
- Nicotine and other toxic components of tobacco smoke can constrict blood vessels, reducing blood flow to the uterus and placenta.

Definitions of Terms

Small Gestational Age

Newborn smaller than usual for the number of weeks of pregnancy.

Low Birth Weight

Newborn weight less than 5.5 pounds or 2,500 grams.

Preterm Birth

Born before 37th weeks of pregnancy has been completed.

Study 1: Parental Smoking During Pregnancy and Its Association with LBW, SGA and Preterm Delivery

June 2005 - July 2006

- A **cohort study**. The population was gathered from the 2005 Taiwan national birth registration data. 90 of the 369 towns were randomly sampled. 24,200 pairs of postpartum women and their newborns were recruited from these towns. (All women were interviewed and the completed interview rate was 87.8%).
- The **exposure** was parental smoking during the different pregnancy stages.
- The **measure of association** was through an interview questionnaire on the number of cigarettes smoked per day. The mothers and fathers were categorized as nonsmoking, 1-10 cigarettes/day, 11-20 cigarettes/day, and >20 cigarettes per day during the three different stages of pregnancy. Including preconception, first trimester, and second and third trimester.
- The **outcome** was the incidence of preterm delivery (gestation < 37 weeks), low birth weight, and small gestational age.
- The **measure of association** was through a comparison of parental smoking and nonsmoking groups through a structured questionnaire six months after their deliveries.

Study 1: Parental Smoking During Pregnancy and Its Association with Preterm Birth Offspring

- **Statistical Analysis:** Multiple Logistic Regression analyses to estimate the odds ratio and 95% confidence intervals of preterm delivery, LBW, and SGA.
- The statistical threshold for significance was set at 0.05.
- Adjusted for potential confounding variables including maternal age, nationality, education, parity, total weight gain during pregnancy, gender of the infant, multifetus, and paternal smoking during the same period.
- A dose-response relationship was tested by trend test.

11-20 cigarettes/day and > 20 cigarettes/day had an odds ratio of 1.20 (0.89-1.60) and 2.78 (1.27-6.10).

Table 5 Parental smoking on preterm birth, low birth weight, and small for gestational age by multiple logistic regression.

Parental smoking	Preterm delivery OR (95% CI)	Low birth weight OR (95% CI)	Small for gestational age OR (95% CI)
Maternal smoking*			
Preconception			
No	Reference	Reference	Reference
1–10 cigarettes/day	1.21 (0.97–1.51)	1.64 (1.31–2.07)	1.35 (1.11–1.64)
11–20 cigarettes/day	1.43 (1.00–2.04)	2.48 (1.76–3.49)	2.05 (1.53–2.77)
>20 cigarettes/day	2.78 (1.27–6.10)	3.46 (1.02–5.97)	0.64 (0.20–2.08)
Test for trend	0.001	<0.0001	<0.0001
First trimester			
No	Reference	Reference	Reference
1–10 cigarettes/day	1.20 (0.89–1.60)	2.27 (1.72–2.99)	2.06 (1.62–2.61)
11–20 cigarettes/day	1.63 (0.89–2.98)	2.72 (1.53–4.85)	2.17 (1.28–3.67)
>20 cigarettes/day	3.58 (0.97–13.30)	3.12 (0.67–14.60)	0.88 (0.11–6.88)
Test for trend	0.013	<0.0001	<0.0001
Second and third trimesters			
No	Reference	Reference	Reference
1–10 cigarettes/day	1.09 (0.80–1.51)	2.25 (1.68–3.02)	2.34 (1.82–3.00)
11–20 cigarettes/day	1.43 (0.67–3.07)	2.71 (1.33–5.49)	2.61 (1.40–4.86)
>20 cigarettes/day	3.87 (1.01–14.82)	6.33 (1.60–25.00)	1.10 (0.14–8.80)
Test for trend	0.092	<0.0001	<0.0001

Main Results

Strengths & Limitations

The proportions of maternal smoking before pregnancy during the first trimester (7.7%), second (3.5%), and third trimester (2.8%) (Table 1).

The offspring of mothers who smoked during pregnancy had lower gestational age and birth weight compared to the offspring of mothers who did not smoke (Table 2).

The incidence of preterm birth, low birth weight, and small gestational age increased in the maternal smoking groups, especially among mothers who smoked more than 20 cigarettes per day (Table 3).

Maternal smoking, especially smoking 1-10 cigarettes and more than 20 cigarettes per day had a significant effect on gestational age (Table 4).

Paternal smoking was not significantly correlated with preterm birth, low birth weight, or small gestational age (Table 5).

Strengths:

- Control for confounding variables
- A large sample size (precise estimates of findings)
- Observational study allows investigators to examine associations over a period of time.

Limitations:

- Validity of smoking information as participants may provide socially desirable responses
- Recall bias on smoking habits during pregnancy with the self-reported data

Study 2: Preterm delivery and exposure to active and passive smoking during pregnancy

October 1999 - September 2000

- A **case-control study**. The population was gathered from nine cities in Italy. Cases were singleton babies before the 37th gestational week and 35th gestational week. Controls were babies with gestational ages greater than or equal to 37th week. A total of 299 preterm cases and 855 controls were analyzed. 96% of eligible mothers participated in the questionnaire.
- The **exposure** was active and passive smoking (Environmental Tobacco Smoking) with the number of people who smoke in the home.
- The **measure of association** was a self-administered questionnaire.
- The **outcome** was preterm delivery (babies born before the 37th gestational week) and early preterm delivery (before the 35th gestational week).
- The **measure of association** was through clinical data about mothers' health at delivery and birth outcomes.

Study 2: Preterm delivery and exposure to active and passive smoking during pregnancy

Statistical Analysis: Univariable and multivariable regression procedures were applied. Estimated Odds Ratio and 95% Confidence Intervals through logistic regression. The P value was set at 0.05 significance

10+ cigarettes/day had an odds ratio of 2.46 (1.05,5.76).

Table 2. Smoking habits of cases (preterm and early preterm delivery) and controls before and during pregnancy				
	Early preterm delivery <35 weeks gestation		Preterm delivery <37 weeks gestation	
	Cases/controls	Adjusted OR ^a [95% CI]	Cases/controls	Adjusted OR ^a [95% CI]
Smoking during pregnancy				
No	80/721	1.00 Reference	239/721	1.00 Reference
Yes	25/125	2.00 [1.16, 3.45]	60/125	1.53 [1.05, 2.21]
Cigarettes/day during pregnancy				
None	80/721	1.00 Reference	237/721	1.00 Reference
<10/day	16/89	1.90 [1.01, 3.56]	40/89	1.54 [1.01, 2.35]
10+/day	9/36	2.46 [1.05, 5.76]	20/36	1.69 [0.91, 3.13]
Smoking before and during pregnancy				
Neither	62/595	1.00 Reference	187/595	1.00 Reference
Yes, only before pregnancy	18/128	1.19 [0.63, 2.24]	50/128	1.20 [0.81, 1.79]
Yes, both before and during pregnancy	25/125	1.43 [1.09, 1.89]	60/125	1.25 [1.04, 1.50]
^a Adjusted for maternal age, previous preterm deliveries, hypertension, diabetes, antenatal class attendance and moderate physical activity.				

Main Results

Table 1 included percentages of specific characteristics that make mothers of preterm and early preterm deliveries more likely to have. These characteristics included the gender of the baby, maternal age, marital status, and working during pregnancy.

There is a strong association between preterm delivery and smoking during pregnancy (Table 2).

ETS exposure is associated with an increase in risk for early preterm delivery for mothers who do not smoke (Table 3).

Women exposed to both active and passive smoking face an elevated risk, particularly in the context of active maternal smoking linked to the increased likelihood of preterm delivery (Table 4).

Strengths & Limitations

Strengths:

- Controlled potential confounding, (maternal age, hypertension, diabetes, moderate physical activity)
- Had a large sample size
- Data on women who reported smoking before pregnancy are similar to the data retrieved from the Italian Institute for Statistical Research on the prevalence of women who smoke

Limitations:

- Coverage of preterm births was limited to 40-60% in four of the nine cities chosen for the study
- Misclassification through the self-report
- Accurate reporting of smoking habits
- Only included Caucasian women, cannot generalize findings for all of the population
- Did not include pregnancies with neonatal malformations
- No temporal sequence

Comparison

Both of these studies are in agreeance that maternal smoking poses a risk for preterm delivery. Additionally, as the number of cigarettes per day increases, so does the severity of adverse effects. The same statistical analyses were used for both articles. Also, both utilized self-reporting data. Both controlled for confounding variables but no effect modifiers were mentioned.

Study 1 has a stronger methodology.

- The larger sample size increases precision. Also, 90 towns were randomly sampled, while the cities in Study 2 were chosen.
- The population in Study 1 did not exclude certain races.
- Study 1 was more recent.
- The likelihood of selection bias increased in Study 2 as diversity sampling did not occur.

Future Direction

Future research on **genetic factors** and **secondhand smoke exposure** on preterm delivery is important.

Genetic factors can influence the metabolism of nicotine and vulnerability to nicotine addiction.

Furthermore, long-term health outcomes for children born to mothers who smoked during pregnancy can provide valuable insights. Lastly, preterm birth is an important cause of neonatal morbidity and mortality and future research must continue.



References

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